

ASSESSMENT COVER SHEET AND TEMPLATE

Section A – to be completed by the student

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Detecting Signals of Natural Selection and Introgression in the European Wildcat (*Felis silvestris*) Using Genome-Wide SNP Data

Abstract

This study explores how natural selection and introgression affect genomic variation in the European wildcat (*Felis silvestris*), a species threatened by hybridisation with domestic cats. Genome-wide SNP data from 108 cats was analysed using the pcadapt package to identify genes disproportionately influencing population structures, giving 1,275 candidate intervals for functional annotation. Eleven were functionally annotated, with seven of them (TRPC4, SLC6A6, SLC6A11, SLC6A1, PDE4B, CACNA1H, and CACNA1E) having strong outlier peaks, allowing further interpretation. Gene ontology (GO) molecular functions were GABA transport, taurine transport, and calcium ion signalling, suggesting that behavioural and metabolic pathways might be under selection. Principal component analyses (PCAs) showed patterns of introgression, local adaptation, and regional divergence. Population history and linkage disequilibrium were limiting factors, but the genes showed plausible genomic regions for changing wildcat ecology. These results may highlight potential markers and offer useful insights for conservation efforts, though further testing would be needed.

Introduction

Anthropogenic hybridisation, human-driven gene flow between wild populations and their domestic relatives, is increasingly recognised as a major conservational challenge (Allendorf *et al.*, 2001; McFarlane & Pemberton, 2019). Introgression is the result of hybridisation, where alleles from one population or species are permanently incorporated into the genome of another through repeated backcrossing (Harrison & Larson, 2014). Hybridisation can occur naturally and sometimes contributes to adaptive evolution (Barton, 2001; Marques *et al.*, 2019). However, the scale and speed of introgression caused by human activity can greatly change the genetic integrity and ecology of threatened species (Todesco *et al.*, 2016; Wayne & Shaffer, 2016).

The European wildcat (*Felis silvestris*) is a well-characterised example of this. Across much of Europe hybridisation with the domestic cat (*Felis catus*) has increased over the past century (Mattucci *et al.*, 2019; Howard-McCombe *et al.*, 2023), and Scottish wildcats now form a near-continuous hybrid swarm (Senn *et al.*, 2019). This likely reflects the recent increase in anthropogenic pressures and feral domestic cats (Quilodrán *et al.*, 2020). Understanding which introgressed variants are spread or removed by selection is essential for assessing the conservation value of hybrids and informing management strategies (Hamilton & Miller, 2016).

Selection can either slow or facilitate gene flow at specific genes depending on the fitness effects of introgressed variants (Daub *et al.*, 2013). In hybridising species, domestic alleles linked to immunity, sensory perception, behaviour, and morphology can introgress at higher or lower rates than the genomic background (Montague *et al.*, 2014). For felids, several traits, including cold tolerance, pathogen resistance, and tameness, differ between wild and domestic forms and offer likely targets for selection (Montague *et al.*, 2014). Introgression for other species, such as the Alpine ibex, shows that domestic immune variants can sometimes increase genetic variation in small populations, highlighting the need to evaluate both costs and benefits of hybridisation (Grossen *et al.*, 2014).

Detecting selection in admixed genomes requires methods that are robust to continuous ancestry. Principal component (PC) based genome scans, from the pcadapt package in RStudio, identify SNPs whose loadings disproportionately drive population structure without requiring previous population assignment (Luu *et al.*, 2017; Privé *et al.*, 2020). Outlier SNPs can then be mapped to nearby genes and assessed for enrichment using functional tools such as g:Profiler, which identify over-represented pathways of gene ontology (GO) categories relevant to physiology, behaviour, or immunity (Kolberg *et al.*, 2023).

Therefore, this project applies these approaches to whole-genome and reduced-representation SNP datasets from wild, domestic, and captive cats to identify genomic regions showing unusual patterns of differentiation or association with hybridisation gradients. Genes near SNP clusters were looked at more closely to see if they had any known functions that might explain patterns of selection or introgression. Based on domestication genomics, variation in neural excitability, inhibitory signalling, and diet-associated metabolism were predicted to show stronger structure where selection and introgression interact.

Methods

Data sources

Whole-genome SNP data for chromosomes 1-18 were provided by Dr. Jo Howard-McCombe and Prof. Mark Beaumont, along with the accompanying metadata describing each cat's population type and geographic origin. These metadata were used to assign individuals to population (Wild, Domestic, Captive) and regional (Scottish, European, Other) categories used throughout downstream analysis (Table 1).

Table 1. Summary of individuals included in the genomic dataset after quality filtering. Counts of captive, domestic, and wild cats sampled across different geographic regions. Two low-quality individuals (SRR2062187 and SRR2062538; Appendix Table A1) were removed as they were non-target species. For downstream population-structure analysis, individuals were grouped into three regional categories: Scottish, European, and Other. The European group included samples from Germany, Belgium, Italy, Spain, Sweden, Switzerland, Portugal, Denmark, Finland, Greece, and the Czech Republic. The Other group included individuals originating from China, Iraq, Jordan, Madagascar, Oman, South Korea, Thailand, and the USA. These regional groupings were used to interpret PCA clustering and genotype-population patterns (total $n = 108$).

Population Type	Country	n
Captive	Belgium	2
Captive	Czech Republic	1
Captive	Finland	1
Captive	Scotland	19
Captive	Sweden	1
Captive	Switzerland	2
Domestic	China	4
Domestic	Denmark	1
Domestic	Iraq	1

Domestic	Italy	1
Domestic	Jordan	1
Domestic	Madagascar	1
Domestic	Oman	1
Domestic	Portugal	1
Domestic	Scotland	6
Domestic	South Korea	1
Domestic	Thailand	1
Domestic	USA	2
Wild	Germany	7
Wild	Greece	2
Wild	Scotland	47
Wild	Spain	2
Wild	Switzerland	3

Variant filtering and genotype processing

To reduce computational load, raw VCF files for all autosomes were filtered using VCFtools, in Unix, and converted to PLINK format. Two individuals of non-target species were removed prior to analysis. SNPs were retained if they had passed a minimum allele count of 3, had no missing genotypes, and did not deviate from Hardy-Weinberg equilibrium ($HWE\ p \geq 1 \times 10^{-4}$). To reduce linkage whilst preserving genome-wide representation, sites were thinned to a minimum spacing of 2,000 bp. Filtered SNPs were exported as PLINK text files and converted to binary PLINK files (.bed/.bim/.fam) with non-standard chromosome support enabled. The sex chromosome was excluded due to ploidy differences between males and females, which violates assumptions of the PCA-based method.

Genome-wide selection scan using pcadapt

Filtered PLINK files were imported into R (v4.4.1), within RStudio, and chromosome-level data were processed using the pcadapt package (Privé *et al.*, 2020). The two non-target individuals (SRR2062187 and SRR2062538; Appendix Table A1) were excluded from all population structure and selection analyses. $K = 5$ was selected from scree plots because these components captured the major structures whilst limiting noise. Pcadapt identifies SNPs that load disproportionately on major PCs, and p-values were merged across chromosomes to generate a Manhattan plot. SNPs with $-\log_{10}(P) > 20$ were classified as outliers and extracted into a combined list using a custom R loop. This threshold minimised false positives under population-structure inflation and ensured that only robust multi-SNP peaks were retained.

Functional annotation and gene identification (g:Profiler, NCBI, Ensembl)

Outlier SNPs identified by pcadapt were converted into 2 kb genomic intervals centred on each SNP position (± 1000 bp). Using genomic intervals instead of individual SNPs ensured that g:Profiler captured any genes located near the candidate SNPs, accounting for annotation uncertainty and linkage to nearby functional regions. The list of genomic regions was submitted to g:Profiler (g:GOST) as query intervals, allowing the identification of any annotated genes located within or overlapping each region. G:Profiler also tested whether the resulting gene set showed enrichment for specific GO molecular functions, pathways, or biological processes, applying the g:SCS multiple-testing correction ($\alpha = 0.05$). Gene identities, coordinates, and functional annotations were then verified manually using NCBI Gene and Ensembl (Felis_catus 9.0) to ensure accurate gene assignment. Confirmed gene names were subsequently added to the Manhattan plot for reference and chromosome mapping.

PCA visualisation of candidate genes

For each candidate gene region, the lead SNP was defined as the variant with the highest absolute pcadapt loading within the annotated window (z-score). Two complementary PCA visualisations were then performed. First, a population-level PCA was generated, plotting individuals along PC1 and the PC most strongly associated with the lead SNP, with points coloured by geographic region (Scottish, European, Other) and shaped by population type (Wild, Domestic, Captive). Then, a SNP-level PCA was produced using the same PC axes, but with individuals coloured by genotype class at the lead SNP (Homozygous A1, Heterozygous, Homozygous A2), allowing visualisation of allele-frequency structure. These analyses enabled the evaluation of how each high-significance SNP aligns with broader population structure and how genotype classes distributed across populations.

Results

After filtering for genotype completeness, allele count, and Hardy–Weinberg equilibrium, 108 individuals remained for analysis (Table 1). Genome-wide pcadapt scores showed substantial variation across autosomes, with multiple SNPs surpassing the outlier threshold ($-\log_{10}(P) > 20$; Figure 1).

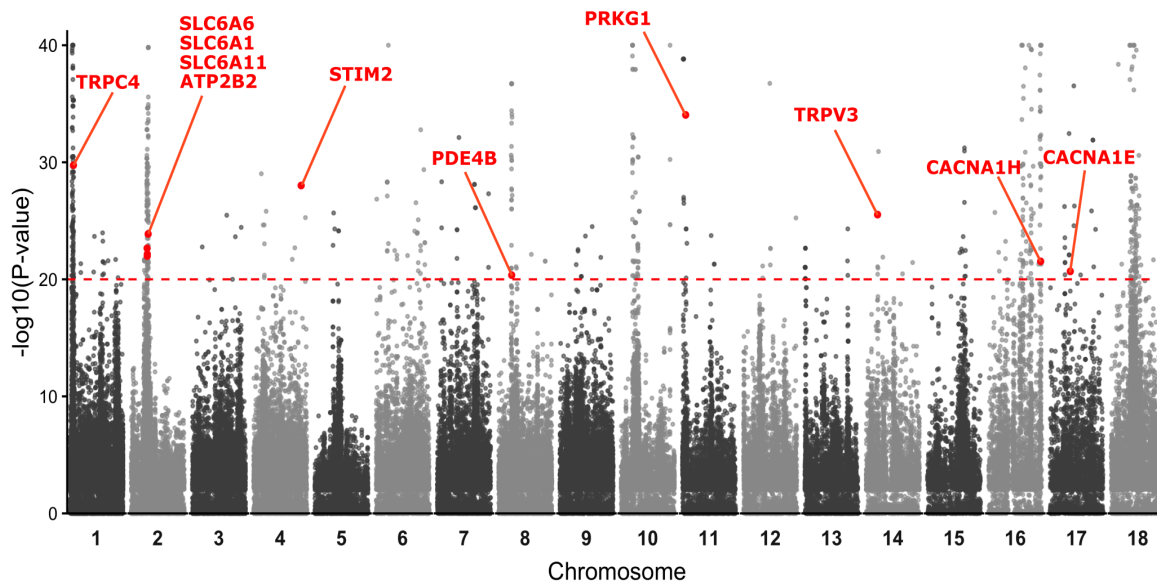


Figure 1. Manhattan plot showing the $-\log_{10}(P)$ values for all SNPs across the genome, generated using *pcadapt* ($K = 5$). Each point represents an individual SNP, ordered by chromosome along the x-axis. The horizontal red dashed line marks the genome-wide significance threshold at $-\log_{10}(P) > 20$. SNPs surpassing this threshold and mapping to annotated genes recovered from *g:Profiler* are highlighted in red and labelled. Eleven genes showed significant association signals: *TRPC4*, *ATP2B2*, *SLC6A6*, *SLC6A1*, *SLC6A11*, *STIM2*, *PDE4B*, *PRKG1*, *TRPV3*, *CACNA1H*, *CACNA1E*. These regions might reflect influences from hybridisation or structure.

These SNPs defined 1,275 genomic intervals of 2 kb ($\pm 1,000$ bp), which were used for functional annotation. *G:Profiler* identified eleven annotated genes overlapping outlier intervals: *TRPC4*, *ATP2B2*, *SLC6A6*, *SLC6A1*, *SLC6A11*, *STIM2*, *PDE4B*, *PRKG1*, *TRPV3*, *CACNA1H*, and *CACNA1E*. Several GO molecular-function terms remained significant after correction, including gamma-aminobutyric acid transport, taurine transmembrane transport, and calcium ion transmembrane transport (Appendix Table A2). For each annotated gene, the SNP with the strongest absolute *pcadapt* loading was designated as the lead SNP (Table 2). Seven genes with the clearest Manhattan-plot signals were visualised using paired PCA analyses (Figure 2): *TRPC4*, *SLC6A6*, *SLC6A1*, *SLC6A11*, *PDE4B*, *CACNA1H*, and *CACNA1E*.

Table 2. Summary of lead SNPs and functional annotations for candidate genes identified from *pcadapt* genome scan. For each gene, the lead SNP corresponds to the variant with the strongest absolute *pcadapt* loading within the gene window. The genomic position (BP), principal component (PC) used on the y-axis of the SNP-level PCA (Figure 2), and

associated z-score are shown. Functional annotations reflect significantly enriched gene ontology (GO) molecular function terms returned by g:Profiler. Together, these values summarise the strength of SNP differentiation and the biological pathways associated with each candidate locus.

Gene	Chromosome	Lead SNP BP	PCA axis (y)	Z-score (y-axis)	Enriched GO Term(s)
TRPC4	CHR 1	16,408,728	PC4	11.25	Calcium ion transport
SLC6A6	CHR 2	55,050,625	PC2	7.73	Taurine transport
SLC6A1	CHR 2	52,199,154	PC4	-9.81	GABA transport
SLC6A11	CHR 2	52,018,022	PC2	6.22	GABA transport
PDE4B	CHR 8	54,689,708	PC4	-11.65	cAMP metabolism
CACNA1H	CHR 16	43,582,578	PC5	8.03	Voltage-gated calcium channel
CACNA1E	CHR 17	28,670,951	PC3	13.17	Voltage-gated calcium channel

In population-level PCAs (Figure 2A–G, left panels), individuals broadly clustered by region (Scottish, European, Other) and population type (Wild, Domestic, Captive). In SNP-level PCAs (Figure 2A–G, right panels), genotype classes formed distinct clusters along the PC most associated with each lead SNP. Z-scores for these axes ranged from -11.65 to 13.17, with CACNA1E showing the highest loading (Table 2). Annotated genes with weaker structure or single-SNP peaks are presented in Appendix Figure A2.

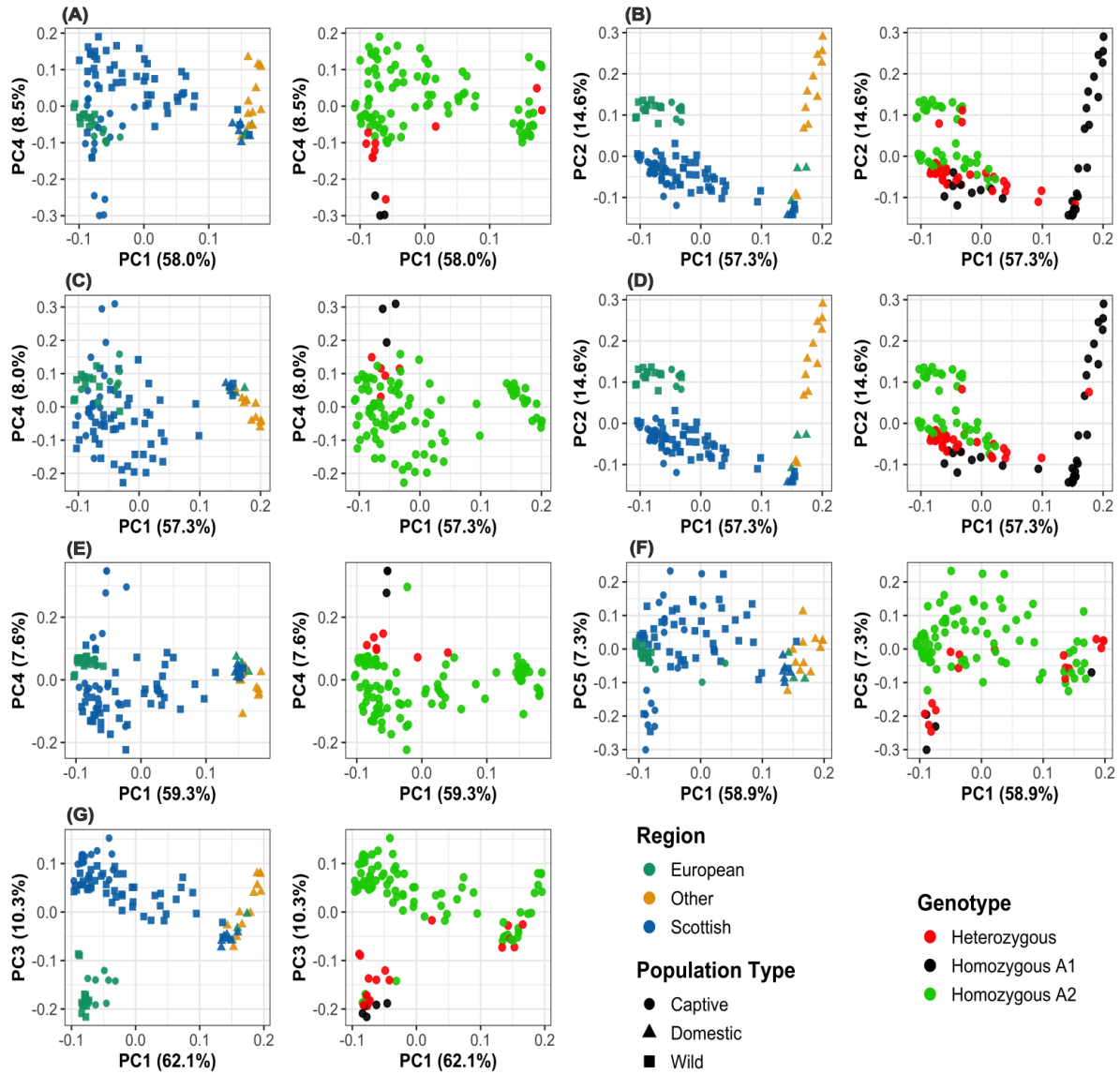


Figure 2. Principal component analyses (PCA) showing population structure and lead-SNP genotype patterns for the seven genes with the strongest *pcadapt* signals ($K = 5$). In each gene panel (A–G), the left plot shows how individuals separate along PC1 and the PC capturing the most SNP-level variation within that gene window. Points are coloured by region (Scottish, European, Other) and shaped by population type (Wild, Domestic, Captive). The right plot shows the matching SNP-level PCA, where individuals are coloured by their genotype at the lead SNP (Homozygous A1, Heterozygous, Homozygous A2). The axis labels show the percentage of variance explained by each PC. Panels correspond to: (A) *TRPC4*, (B) *SLC6A11*, (C) *SLC6A1*, (D) *SLC6A6*, (E) *PDE4B*, (F) *CACNA1H*, and (G) *CACNA1E*. Additional annotated genes are presented in Appendix Figure A1.

Discussion

To interpret pcadapt outliers, each locus was evaluated in the context of allele-frequency structure and gene function. Because Scottish wildcats form a bottlenecked, structured lineage, several outlier peaks, particularly single-SNP signals, may reflect demographic history rather than selection. The g:Profiler query identified eleven annotated outlier genes, with seven showing the strongest interpretable signals.

TRPC4 encodes receptor-controlled Ca^{2+} channels and contributes to GPCR-dependent Ca^{2+} signalling in central neurons, with potential impacts on neuronal excitability (Plant & Schaefer, 2003). Allele patterns at TRPC4 (Figure 2A) show A2 is common across populations but has strong representation in domestic cats, whilst the A1 allele is mostly confined to Scottish and other wild individuals. *Trpc4*-knockout mice exhibit reduced innate fear responses, but the link between specific TRPC4 genotypes and behavioural phenotypes remains poorly understood (Riccio *et al.*, 2014). Thus, the presence of the domestic-associated A2 allele in Scottish wildcats is consistent with introgression, but interpretations should remain cautious, as the signal suggests modulation of Ca^{2+} -dependent excitability rather than direct evidence of behavioural change.

SLC6A1 and SLC6A11 encode the GABA transporters GAT1 and GAT3, which are responsible for terminating inhibitory synaptic signalling (Andersen, 2025). The A1 allele of SLC6A11 is largely confined to Scottish wildcats and captive individuals (Figure 2B), indicating a regional restriction that may reflect local adaptation or retained ancestral diversity in structured populations (Savolainen *et al.*, 2013; Slatkin, 1987). The mixed allele distribution of SLC6A1 across populations (Figure 2C) suggests shared ancestral variation or introgression. Since GABAergic pathways modulate fear and social behaviour, variation at these regions could influence behavioural differences, although phenotypic effects in felids remain untested (Liu *et al.*, 2007).

SLC6A6 encodes TauT, which moves taurine across cell membranes; disabling this gene in mice can lead to retinal degeneration (Preising *et al.*, 2019). The SLC6A6 A1 allele is present across all population clusters (Figure 2D). Although taurine physiology has not been studied directly in wildcats, prey-switching towards taurine-poor species, such as rabbits, could alter exposure to dietary taurine, particularly in human-provided landscapes where prey availability differs (Owens *et al.*, 2021). If domestic-derived A1 alleles enhance taurine uptake or increase tolerance to variable dietary taurine, they may introgress into wild and captive populations exposed to anthropogenic feeding. Widespread distributions of the A1

allele across population types may therefore reflect both introgression and ecological variation in taurine availability.

PDE4B encodes a cAMP-specific phosphodiesterase with key roles in neuronal signalling, synaptic plasticity, and stress-related behaviour in mammals (Delhay & Bardoni, 2021). Rutten *et al.* (2011) demonstrated influence on spatial memory and anxiety-related responses in rodents. The PDE4B A1 allele is found exclusively in Scottish wildcats and captive individuals and is absent from domestic cats and continental wildcats (Figure 2E), indicating regional adaptation or retained ancestral wildcat diversity rather than introgression. Whilst PDE4B has not been directly implicated in selection in other domestic species, such as rabbits or goats, its broader functional role makes it a candidate for further investigation in the context of wild adaptation (Carneiro *et al.*, 2014; Zheng *et al.*, 2020).

CACNA1H and CACNA1E encode T-type and R-type calcium channels involved in neuronal firing thresholds and behavioural responsiveness (Zamponi *et al.*, 2015). The A1 alleles predominate in European and Other wildcats, whereas Scottish wildcats and domestic cats are enriched for A2 (Figure 2F–G); consistent with reduced diversity and behavioural stabilisation during domestication (Larson & Fuller, 2014; Montague *et al.*, 2014). These differences could be a result of historical isolation or drift, with regional structure also affecting results, but this would need further analysis (Driscoll *et al.*, 2007). Interestingly, the calcium-channel findings might reflect demographic history rather than true selection. This makes it more difficult to determine genetic drift from adaptation.

Similar patterns appear in other hybridising mammals. Wolves hybridising with dogs shows behavioural differences such as stress reactivity and sociability likely reflecting domestication-linked traits, though evidence for introgression of such alleles into wild populations is limited (Range & Marshall-Pescini, 2022). For Alpine ibex, goat-derived immune variants increase diversity in small herds (Grossen *et al.*, 2014). These examples show that introgression often affects neural, sensory, or immune pathways, consistent with these findings. Across these genes, an ecological pattern shows that variation in inhibitory signalling, calcium-dependent excitability, and taurine metabolism aligns with traits like vigilance, hunting, and metabolic efficiency. Because these functions differ between wild and domestic felids, these patterns could be a mix of introgression and historical effects, but certainty is difficult (Montague *et al.*, 2014)

Several limitations constrain interpretation in this study. Pcadapt outliers are best seen as descriptive signals, where they can't easily separate selection from historical drift vents,

limiting certainty (Hoban *et al.*, 2016). Loadings may therefore reflect shared drift or historical bottlenecks rather than true selective sweeps (Pavlidis *et al.*, 2012; Lotterhos & Whitlock, 2014). Furthermore, PCs do not quantify allele frequency divergence between groups. Metrics such as F_{st} are better suited for that purpose but was not applied here as the data is not discrete (Lotterhos & Whitlock, 2014). Future analyses could incorporate F_{st} with clearly defined group structures.

Linkage disequilibrium (LD) was not accounted for in this analysis. A single selection event can affect multiple nearby SNPs, inflating the appearance of distinct outlier peaks (Nielsen *et al.*, 2005; Pavlidis *et al.*, 2012). Without LD pruning, some high-significance clusters likely reflect shared haplotypes rather than independent selective events (Daub *et al.*, 2013). To distinguish single selective sweeps from blocks of correlated SNPs, haplotype-based scans such as iHS or XP-EHH could be applied within candidate regions to identify unusually long, high frequency haplotypes expected under recent selection (Voight *et al.*, 2006; Sabeti *et al.*, 2007). Incorporating haplotype-based or admixture-aware models would substantially improve inference beyond PCA-based outlier detection.

The $-\log_{10}(P) > 20$ threshold was arbitrary, so some true positives just below it may have been excluded, whilst high peaks within LD blocks may inflate false positives. This directly influences which functional pathways appear enriched and may partly explain the absence of immune gene regions, such as MHC, which are known to be influenced by introgression in wildcats (Howard-McCombe *et al.*, 2023). Although focusing on multi-SNP peaks reduced noise, enrichment results should still be interpreted cautiously, given the limitations of annotation tools, including pathway bias (Hoban *et al.*, 2016; Huang *et al.*, 2009).

Whilst genome scans are effective at detecting large differences, smaller details can be missed, especially in genomic systems involving many genes; behavioural and metabolic traits likely reflect many small shifts that pcadapt cannot determine without further analysis (Pritchard *et al.*, 2010; Berg & Coop, 2014). Additionally, the “omnigenic hypothesis” states variants outside core pathways may still shape complex traits through subtle regulatory interactions (Boyle *et al.*, 2017). Thus, whilst the highlighted genes provide strong candidate targets for selection, they likely represent only a subset of the genes contributing to domestication and introgression.

Conclusion

This study identifies genomic regions where introgression and population history may influence wildcat behavioural and metabolic traits. These results highlight markers with

immediate conservation value. Domestic-associated variants at TRPC4 and CACNA1E can guide hybrid detection, whereas regionally restricted alleles like PDE4B A1 may warrant targeted preservation. This might help inform future breeding programs, especially if the goal is to keep certain wildcat traits intact. Continued control of feral domestic cats remains essential to limit gene flow, particularly in heavily admixed Scottish populations. Future research should assess fitness effects of introgressed alleles to guide genomically informed conservation planning for *Felis silvestris*.

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Appendix

Table A1. Sample identifiers, putative species assignments, source countries, and population types are shown for all individuals originally compiled for analysis. The two individuals SRR2062187 and SRR2062538 were excluded from downstream genomic analyses because they are non-target *Felis* species and lacked accompanying source country and population metadata. They are retained here for completeness.

Sample ID	Putative Species	Source Country	Source Population
FA661_E1	<i>Felis silvestris</i>	Germany	Wild
SRR2062187	<i>Felis chaus</i>	N/A	N/A
SRR2062538	<i>Felis margarita</i>	N/A	N/A
SRR5040107	<i>Felis catus</i>	Oman	Domestic
SRR5040110	<i>Felis catus</i>	USA	Domestic
SRR5040111	<i>Felis catus</i>	Madagascar	Domestic
SRR5040113	<i>Felis catus</i>	Iraq	Domestic
SRR5040114	<i>Felis catus</i>	Portugal	Domestic
SRR5040116	<i>Felis catus</i>	Denmark	Domestic
SRR5040117	<i>Felis catus</i>	Italy	Domestic
SRR5040118	<i>Felis catus</i>	Jordan	Domestic
SRR5040120	<i>Felis catus</i>	Thailand	Domestic
SRR5040125	<i>Felis catus</i>	USA	Domestic
SRR5040126	<i>Felis catus</i>	South Korea	Domestic
SRR7621212	<i>Felis catus</i>	China	Domestic
SRR7621222	<i>Felis catus</i>	China	Domestic
SRR7621225	<i>Felis catus</i>	China	Domestic
SRR7621252	<i>Felis catus</i>	China	Domestic
WCQ0008	<i>Felis silvestris</i>	Scotland	Wild
WCQ0010	<i>Felis silvestris</i>	Scotland	Wild
WCQ0020	<i>Felis silvestris</i>	Scotland	Wild
WCQ0034	<i>Felis silvestris</i>	Scotland	Wild
WCQ0040	<i>Felis silvestris</i>	Scotland	Wild
WCQ0053	<i>Felis silvestris</i>	Scotland	Wild
WCQ0058	<i>Felis silvestris</i>	Scotland	Wild
WCQ0059	<i>Felis silvestris</i>	Scotland	Wild
WCQ0072	<i>Felis silvestris</i>	Scotland	Wild
WCQ0080	<i>Felis silvestris</i>	Scotland	Wild
WCQ0086	<i>Felis silvestris</i>	Scotland	Wild
WCQ0105	<i>Felis silvestris</i>	Scotland	Wild
WCQ0107	<i>Felis silvestris</i>	Scotland	Wild
WCQ0117	<i>Felis silvestris</i>	Scotland	Wild
WCQ0119	<i>Felis silvestris</i>	Scotland	Wild
WCQ0124	<i>Felis silvestris</i>	Scotland	Wild
WCQ0127	<i>Felis silvestris</i>	Scotland	Wild
WCQ0131	<i>Felis silvestris</i>	Scotland	Wild

WCQ0146	Felis silvestris	Scotland	Wild
WCQ0148	Felis silvestris	Scotland	Wild
WCQ0159	Felis silvestris	Scotland	Wild
WCQ0217	Felis silvestris	Scotland	Captive
WCQ0244	Felis silvestris	Scotland	Captive
WCQ0245	Felis silvestris	Scotland	Captive
WCQ0341	Felis silvestris	Scotland	Captive
WCQ0345	Felis silvestris	Scotland	Captive
WCQ0387	Felis silvestris	Scotland	Wild
WCQ0399	Felis silvestris	Scotland	Captive
WCQ0400	Felis silvestris	Scotland	Captive
WCQ047	Felis silvestris	Scotland	Wild
WCQ052	Felis silvestris	Scotland	Wild
WCQ0606	Felis silvestris	Scotland	Wild
WCQ0610	Felis silvestris	Scotland	Captive
WCQ0614	Felis silvestris	Scotland	Captive
WCQ0646	Felis silvestris	Scotland	Wild
WCQ0665	Felis silvestris	Scotland	Wild
WCQ0888	Felis silvestris	Scotland	Wild
WCQ0925	Felis silvestris	Scotland	Wild
WCQ0928	Felis silvestris	Scotland	Wild
WCQ0937	Felis silvestris	Scotland	Wild
WCQ099	Felis silvestris	Scotland	Wild
WCQ100	Felis silvestris	Scotland	Wild
WCQ1110	Felis silvestris	Finland	Captive
WCQ1111	Felis silvestris	Sweden	Captive
WCQ1133	Felis silvestris	Scotland	Wild
WCQ1135	Felis catus	Scotland	Domestic
WCQ1136	Felis catus	Scotland	Domestic
WCQ1137	Felis catus	Scotland	Domestic
WCQ1138	Felis catus	Scotland	Domestic
WCQ1139	Felis catus	Scotland	Domestic
WCQ1141	Felis silvestris	Scotland	Wild
WCQ1152	Felis silvestris	Scotland	Wild
WCQ1196	Felis silvestris	Scotland	Wild
WCQ1199	Felis silvestris	Scotland	Wild
WCQ1227	Felis silvestris	Scotland	Wild
WCQ1230	Felis silvestris	Scotland	Captive
WCQ1250	Felis silvestris	Czech Republic	Captive
WCQ1316	Felis silvestris	Belgium	Captive
WCQ1317	Felis silvestris	Belgium	Captive
WCQ1324	Felis silvestris	Switzerland	Captive
WCQ1325	Felis silvestris	Switzerland	Captive
WCQ1329	Felis silvestris	Scotland	Captive

WCQ1452	Felis silvestris	Scotland	Captive
WCQ1501	Felis silvestris	Greece	Wild
WCQ1532	Felis silvestris	Greece	Wild
WCQ1561	Felis silvestris	Germany	Wild
WCQ158	Felis silvestris	Scotland	Wild
WCQ1633	Felis silvestris	Spain	Wild
WCQ1634	Felis silvestris	Spain	Wild
WCQ165	Felis silvestris	Scotland	Wild
WCQ168	Felis silvestris	Scotland	Wild
WCQ209	Felis silvestris	Scotland	Wild
WCQ211	Felis silvestris	Scotland	Wild
WCQ212	Felis silvestris	Scotland	Wild
WCQ243	Felis silvestris	Scotland	Captive
WCQ248	Felis silvestris	Scotland	Wild
WCQ343	Felis silvestris	Scotland	Captive
WCQ344	Felis silvestris	Scotland	Captive
WCQ402	Felis silvestris	Scotland	Captive
WCQ427	Felis silvestris	Scotland	Captive
WCQ428	Felis silvestris	Scotland	Captive
WCQ443	Felis catus	Scotland	Domestic
WCQ515	Felis silvestris	Scotland	Wild
WCQ553	Felis silvestris	Scotland	Captive
WK028	Felis silvestris	Switzerland	Wild
WK032	Felis silvestris	Switzerland	Wild
WK145	Felis silvestris	Switzerland	Wild
ex19_E1	Felis silvestris	Germany	Wild
ex21_E1	Felis silvestris	Germany	Wild
ex38_E1	Felis silvestris	Germany	Wild
ex40_E1	Felis silvestris	Germany	Wild
ex9_E1	Felis silvestris	Germany	Wild

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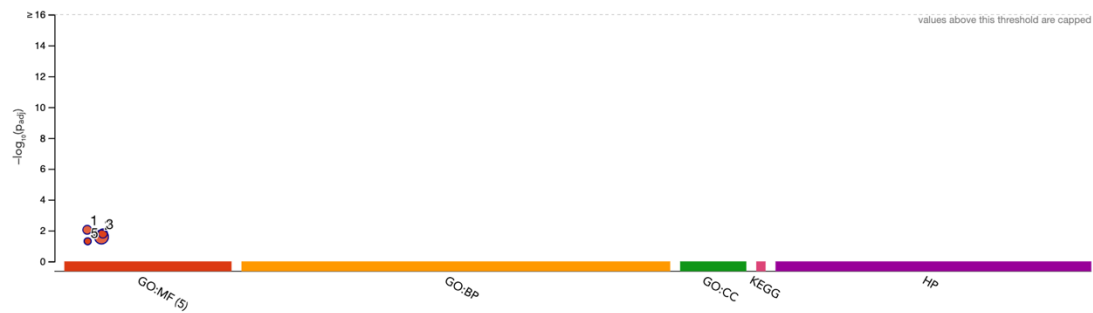
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ID	Source	Term ID	Term Name	P _{adj} (query_1)
1	GO:MF	GO:0005332	gamma-aminobutyric acid:sodium:chloride sy...	8.941×10 ⁻³
2	GO:MF	GO:0015085	calcium ion transmembrane transporter activity	2.608×10 ⁻²
3	GO:MF	GO:0015185	gamma-aminobutyric acid transmembrane tran...	1.554×10 ⁻²
4	GO:MF	GO:0005369	taurine:sodium symporter activity	4.946×10 ⁻²
5	GO:MF	GO:0005368	taurine transmembrane transporter activity	4.946×10 ⁻²

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g:Profiler

Figure A1. Functional enrichment analysis of genes, identified from the 1,275 pcadapt, candidate regions was performed using g:Profiler (g:GOST). The plot shows the distribution of significantly enriched Gene Ontology (GO) terms across molecular function (GO:MF), biological process (GO:BP), cellular component (GO:CC), as well as KEGG and HP categories. The accompanying table lists all GO:MF terms that remained significant after multiple-testing correction, highlighting enrichment related to γ -aminobutyric acid transport, calcium ion transmembrane transport, and taurine-associated transporter activity. Significance values (P_{adj}) are shown as reported by g:Profiler.

Table A2. Significantly enriched gene ontology (GO) molecular-function terms identified by g:Profiler for genomic intervals surrounding outlier SNPs. Table reports GO term name, GO ID, adjusted P-value (P_{adj}), annotated genes within each term, and broader functional category.

GO Term (Name)	GO ID	Adjusted P-value (P _{adj})	Genes Annotated	Functional Category
Gamma-aminobutyric acid:sodium:chloride symporter activity	GO:0005332	8.94x10 ⁻³	SLC6A11, SLC6A1, SLC6A6	GABAergic neurotransmission
Gamma-aminobutyric acid transmembrane transporter activity	GO:0015185	1.55x10 ⁻²	SLC6A11, SLC6A1, SLC6A6	GABAergic neurotransmission
Calcium ion transmembrane transporter activity	GO:0015085	2.61x10 ⁻²	TRPC4, ATP2B2, STIM2, PDE4D, PRKG1, TRPV3, CACNA1H, CACNA1E	Calcium signalling / ion transport
Taurine transmembrane transporter activity	GO:0005368	4.95x10 ⁻²	SLC6A11, SLC6A6	Taurine metabolism / osmoregulation
Taurine:sodium symporter activity	GO:0005369	4.95x10 ⁻²	SLC6A11, SLC6A6	Taurine metabolism / osmoregulation



Figure A2. Principal component analyses (PCAs) for all genes identified via g:Profiler, using genomic intervals generated from significant padapt SNPs. For each gene, the PCA shown corresponds to the SNP with the highest loading within its candidate region. In the left panels (A-G), individuals are coloured by region and population type, with genotype classes in the right panels (A-G). Gene name, chromosome, and base-pair position of the focal SNP are indicated above each panel.